Challenge: Trapped in Time

Team: Pluto

Narrative

INTERCEPTED TRANSMISSION — CELESTIAL VAULT SECURE RELAY — SIGNAL ORIGIN: CLASSIFIED

`At an undisclosed hour, an anomalous signal began propagating through the Celestial Vault FTP relay. The last known operator - callsign PLUTO - left behind a series of encrypted transmissions before going dark. The files are still there, waiting. Someone has embedded a secret message deep within them, hidden beneath layers of steganographic misdirection. Your team has been activated to intercept, decrypt, and extract that message before the window closes. The password changes every 60 seconds. You had better be ready.`

Objective

The goal of this challenge is to login to **THE CELESTIAL VAULT FTP Server** (on standard port 21 — other network details will be provided at the beginning of the challenge) using the following credentials:

`Username:` `pluto`

`Password:` `plutoXXXXY`

XXXXY in the password is a TimeLock code as defined by our TimeLock algorithm... almost. There is an extra character at the end that is simply the last character of the final hash. Of course, the password to the pluto account changes every 60 seconds! How exciting!

Once logged in, you will need to fetch a few files that can somehow be further inspected to reveal one or more messages cleverly embedded within. You are provided no other details, other than that an *exorcism* may need to be performed, followed by closely inspecting bits and bytes.

Important Notes

- At the start of the challenge, you will be provided with two encoded files. Inspect them carefully — the information you need to get started may not be where you expect it to be.
- The current time of the machine implementing the TimeLock algorithm for the pluto user is the actual current local time. Feel free to confirm via:

```
nc <ipaddress> 54321    # Ctrl+C to end connection
```

It should send you back a string with the current time on the server.

Hints

- Once on the FTP server, the following FTP commands may be useful: `passive`, `binary`, `prompt`, `mget *`.
- Once you determine what to do with the files on the FTP server (and get a result), there are actually TWO hidden files in the result (one may be somewhat obvious; the other, not so much).
- For the first hidden file, the offset is a power of 2 (and happens to start with that digit too), and the interval is a power of 2 (note that the BYTE method may work quite well).
- For the second hidden file, the offset is a power of 2, plus 1, and the interval is a power of 2 that is less than or equal to 4 (note that the BIT method may work quite well).
- The following bytes most likely "signal" something: `0x00`, `0xFF`, `0x00`, `0x00`, `0xFF`, `0x00`.
- What's a JFIF? And is it perhaps involved with the BYTE method?
- It's possible that a THIRD hidden file exists inside of one of the two hidden files. Egads! It may be pretty obvious, though. Counting is your friend.

Completion

When you think you have successfully completed the challenge, send the secret passphrase to your team text channel and **@prof**.

Deliverable

Submit a compressed folder to Canvas containing:

1. A document providing a thorough summary of your team's completion of the challenge; include all relevant Steg settings (e.g., `-B -o 1024 -i 8 -w some_file.bmp`) for every stegged document; include screenshots as appropriate; clearly state the final secret passphrase; each team member must contribute their own paragraph describing their role.
2. The "life alert bracelet" image (with its appropriate file extension).
3. The "cyber" image (with its appropriate file extension).
4. The "Constitution" text (with its appropriate file extension).

5. Any modified Golang code (TimeLock, XOR, Steg programs).
6. Any other scripts or programs you created (if applicable).

Grading

You will be graded on how well you:

- Implement the TimeLock algorithm to access the FTP server and obtain the initial files (25%);
- Decrypt the initial documents to obtain the "life alert bracelet" image (10%);
- Determine the Steg details for obtaining the "cyber" image (20%);
- Determine the Steg details for obtaining the "Constitution" text (20%);
- Determine the Steg details for obtaining the final secret passphrase (20%); and
- Thoroughly detail your team's path to complete the challenge, document each member's role, and accurately specify the final secret passphrase (5%).

GOOD LUCK!